

Program	Bachelor of Technology (B.Tech.)	Semester - 5
Type of Course	Core Courses	
Prerequisite	03070301-T - DATA AND FILE STRUCTURE	
Rationale	To teach paradigms and approaches used to analyze and design algorithms and to appreciate the impact of algorithm design To explain different computational models (e.g., divide-and-conquer), order notation and various complexity measures (e.g., running time, disk space) to analyze the complexity/performance of different algorithms.	
Effective From A.Y.	2024-25	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				T	T	P	P	
3	1	0	4	70	30	-	-	100

SEE - Semester End Examination, T - Internal Theory, P - Internal Practical

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	BASICS OF ALGORITHMS AND MATHEMATICS What is an algorithm? Mathematics for algorithmic, Sets, Functions and relations. Vectors and matrices. Linear inequalities and linear equations and master method The efficient algorithm, Average and worst case analysis. Elementary operation, Asymptotic notation, Analyzing control statement, Amortized analysis, Sorting algorithm, Binary tree search.	10	15
2	DIVIDE AND CONQUER ALGORITHM Introduction about divide and conquer Multiplying large integer's problem. Problem solving using divide and conquer algorithm. Binary search, Sorting (Merge Sort, Quick Sort).Matrix multiplication, Exponential.	15	20
3	GREEDY ALGORITHM General characteristics of greedy algorithms.Problem solving using greedy algorithm. Activity selection problem. Elements of greedy strategy, Minimum spanning trees (Kruskal's algorithm, Prim's algorithm). Graphs: Shortest paths, The knapsack problem. Job scheduling problem.	15	20
4	DYNAMIC PROGRAMMING & EXPLORING GRAPHS Introduction, The principle of optimality, Problem solving using dynamic programming. Calculating the binomial coefficient, Making change problem. Assembly line-scheduling, Knapsack problem, Shortest path, Matrix chain multiplication. Longest common Subsequence.	10	25
5	STRING MATCHING & INTRODUCTION TO NP-COMPLETENESS The naive string matching algorithm. The Rabin-Karp algorithm. String matching with finite automata. The class P and NP. Polynomial reduction, NP- Completeness problem, NP-Hard problems.	10	20
Total		60	100

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy						
Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	5	25	20	25	20	5

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:

CO1	Analyze the asymptotic performance of algorithms.
CO2	Derive and solve recurrences describing the performance of divide-and-conquer algorithms.
CO3	Find optimal solution by applying various methods.
CO4	Apply pattern matching algorithms to find particular pattern.
CO5	Differentiate polynomial and nonpolynomial problems.

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PO - 1	2	2	3	3	2
PO - 2	2	3	3	2	2
PO - 3	2	2	2	2	3
PO - 4	2	2	2	3	3
PO - 5	2	1	2	2	2
PO - 6	2	3	2	2	3
PO - 7	3	2	2	3	2
PO - 8					
PO - 9	2	3	2	3	2
PO - 10	3	3	2	1	3
PO - 11	2	2	2	3	2
PO - 12	2	2	2	2	2

CO PSO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PSO - 1	3	3	3	3	3
PSO - 2	2	2	2	2	2
PSO - 3	1	1	1	1	1

Reference Books

1.	Introduction to Algorithms (TextBook) By Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein PHI
2.	Introduction to Design and Analysis of Algorithms By Anany Levitin Pearson